How to open the DRMS in a web browser

Disaster Mapping & Relief Management System · step-by-step guide for a Windows PC with WampServer

> **In short:** double-click `start_localhost.bat`. Wait about 10 seconds. Your browser opens **http://localhost/disaster_and_relief_management/**. On the first visit, create the administrator account and sign in.

1. What you need

- **WampServer** installed in `C:\wamp64` (the scripts use Apache 2.4.58, MySQL 8.2.0 and PHP 8.2.13 from that folder). Requirements of the system: PHP 8.1+ with `mysqli` and `mbstring`, MySQL 8.0.16+.
- Any web browser (Edge, Chrome, Firefox).
- Internet only for the **map** (OpenStreetMap tiles). Everything else works offline.

2. Method A: one click (recommended)

1. Open the project folder. You will see `start_localhost.bat`, `stop_localhost.bat` and the folder `disaster_and_relief_management`.
2. Double-click `start_localhost.bat`. No administrator rights are needed. It starts MySQL and Apache, links the project into `C:\wamp64\www\` the first time, waits 8 seconds and opens the browser.
3. If the page does not open by itself, type this in the address bar: **http://localhost/disaster_and_relief_management/**
4. To stop everything later, double-click `stop_localhost.bat`.

You can run `start_localhost.bat` again at any time: programs that are already running are not started twice. After the PC restarts, run it again.

3. Method B: without Apache

1. Double-click `start_server.bat`. It starts MySQL and PHP's built-in web server and opens **http://127.0.0.1:8099/index.php**.
2. Keep the black command window open while you use the system. Close it to stop the web server (MySQL keeps running).

4. Method C: WampServer by hand

1. Start WampServer and wait until its icon is **green**.
2. Copy the whole `disaster_and_relief_management` folder into `C:\wamp64\www\`.
3. If your MySQL password is not empty, edit `config.php` (host, user, password, port). Keep `name` as `disaster_and_relief_management`.
4. Open **http://localhost/disaster_and_relief_management/**.

5. The first visit

1. The database, all 22 tables, the 8 triggers and the sample data are created **automatically**. You do not import anything in phpMyAdmin.
2. The first-run screen asks you to create the **administrator account**. Choose a username and a password of at least 8 characters.
3. Sign in with that account.

Role	What it can do
admin	Everything: all tables, delete disasters, the Users page
editor	View everything; add, edit and delete records (not whole disasters)
viewer	Read-only, search, CSV export

Five wrong passwords lock an account for 5 minutes. A session ends after 4 hours without a click.

6. What to look at

Page	What you see
Dashboard	Totals, live status cards, “needs attention” list, risk and shelter bars, index of all tables
Live Report	Control-room view that refreshes itself every 5 seconds: warning signals, live disaster path on a map, shelter space, relief distribution
Map	All areas (coloured by risk) and shelters (green / orange / red by how full they are)
Data Health	17 checks that find records that contradict each other
Disaster Report	One printable report per disaster
Search	One box that searches every table
Table pages	Sort, search, page, export CSV, add / edit / delete. Delete shows what else would disappear
Users / Audit Log	Accounts and roles (admin) · who changed what, with old and new values

> The sample data has 13 disasters, 21 shelters, 60 families and 12 organizations (all fictional). Two disasters are **Active**: a flood and a cyclone. Open the **Live Report** first to see them.

7. How to view the ER diagram in a browser

The ER diagram is a PDF file in the project folder, not a page of the web system, so it has no `localhost` address. Open it in a browser like this:

1. Right-click `DRMS_ER_Diagram.pdf` → **Open with** → Microsoft Edge (or Chrome). Or drag the file into an open browser window.
2. Page 1 is the diagram: 22 tables, 29 foreign keys. Page 2 lists the foreign keys, unique keys, CHECK rules and triggers.
3. Zoom with `Ctrl` + mouse wheel. The diagram is large, so zoom in to read the column names.

Other diagram files in the same folder: `DRMS_ER_Diagram.png` (page 1 as a picture), `DRMS_ER_Diagram_Simple.pdf` (9 main tables) and `DRMS_Conceptual_ER_Diagram.pdf` (named relationships).

8. If something goes wrong

Problem	What to do
Apache does not start; `C:\wamp64\logs\apache_error.log` mentions *VCRUNTIME140.dll*	Copy `C:\wamp64\bin\php\php8.2.13\vcruntime140.dll` into `C:\wamp64\bin\apache\apache2.4.58\bin\` and start again (or install the current Microsoft Visual C++ 2015-2022 Redistributable x64).
Cannot reach MySQL	Run `start_localhost.bat` (or `start_server.bat`) again; check host and port in `config.php`.
MySQL refused the user name / password	Fix `user` and `pass` in `config.php`.
Page not found after a restart	Apache and MySQL do not start by themselves: run `start_localhost.bat` again.
Map is grey or says “Access blocked”	No internet, or the browser blocks the Referer header (for example Brave Shields). Tables and lists under the map still work.
Your WampServer has other versions	Edit the paths at the top of `start_localhost.bat` / `start_server.bat`.
Start again from scratch	Drop the database `disaster_and_relief_management` in phpMyAdmin and reload the page. Create the administrator again.

Details and the full feature list: `disaster_and_relief_management\README.md`. This system is for a local PC over plain HTTP; use HTTPS and a limited MySQL user (`create_db_user.sql`) before opening it to a network.